

# Automotive Manufacture Practical Report

```
create database Automotive;
```

```
use Automotive;
```

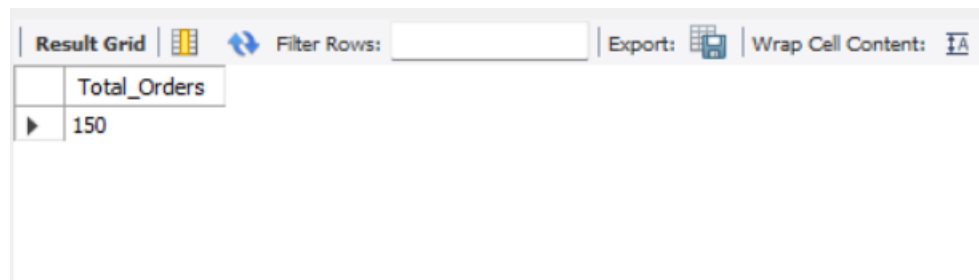
```
select * from automotive_manufacture;
```

```
select * from supplier_details;
```

#1. Find the total number of orders.

```
select count(*) AS Total_Orders
```

```
from automotive_manufacture;
```



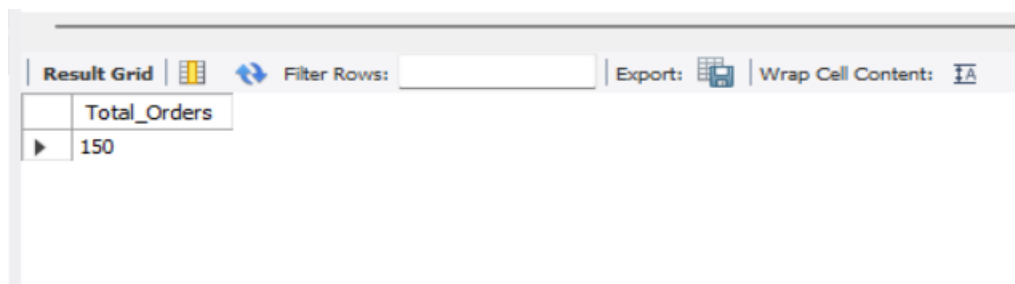
The screenshot shows a database query result grid. The grid has two columns: 'Total\_Orders' and a value '150'. The grid is displayed in a window with a toolbar at the top containing icons for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'.

|   | Total_Orders |
|---|--------------|
| ▶ | 150          |

#2. Find the total number of different automotive parts.

```
select count(distinct `Part_Name`) AS Total_Automotive_Part
```

```
from automotive_manufacture;
```



The screenshot shows a database query result grid. The grid has two columns: 'Total\_Orders' and a value '150'. The grid is displayed in a window with a toolbar at the top containing icons for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'.

|   | Total_Orders |
|---|--------------|
| ▶ | 150          |

#3. Find the total Ordered Quantity.

```
select sum(`Ordered_Qty`) AS Total_Quantity
```

```
from automotive_manufacture;
```

| Result Grid |                       | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-----------------------|--------------|---------|--------------------|
|             | Total_Automotive_Part |              |         |                    |
| ▶           | 12                    |              |         |                    |

#4.Find the total Cost.

```
select sum(`Total_Cost`) AS Total_Cost
from automotive_manufacture;
```

| Result Grid |            | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|------------|--------------|---------|--------------------|
|             | Total_Cost |              |         |                    |
| ▶           | 36005665   |              |         |                    |

#5.Find the average Unit Cost.

```
select avg(`Unit_Cost`) AS Avg_Cost
from automotive_manufacture;
```

| Result Grid |          | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|----------|--------------|---------|--------------------|
|             | Avg_Cost |              |         |                    |
| ▶           | 222.3000 |              |         |                    |

#6.Find the maximum & minimum Unit Cost.

```
select max(`Unit_Cost`) AS Maximum_cost,
min(`Unit_Cost`) AS Minimum_cost
from automotive_manufacture;
```

| Result Grid |              | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|--------------|--------------|---------|--------------------|
|             | Maximum_cost | Minimum_cost |         |                    |
| ▶           | 600          | 45           |         |                    |

#7.Find the number of orders for each supplier.

```

select Supplier_Name, count(*) AS Number_Orders
from automotive_manufacture
group by Supplier_Name;

```

| Result Grid |                      |               | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|----------------------|---------------|--------------|---------|--------------------|
|             | Supplier_Name        | Number_Orders |              |         |                    |
| ▶           | Prime Components     | 22            |              |         |                    |
|             | DriveLine Systems    | 14            |              |         |                    |
|             | Global Motors Parts  | 21            |              |         |                    |
|             | Metro Auto Supply    | 19            |              |         |                    |
|             | Elite Automotive     | 15            |              |         |                    |
|             | SteelTech Components | 18            |              |         |                    |

#8. Find the total Ordered Quantity by supplier.

```

select Supplier_Name, sum(`Ordered_Qty`)
AS Total_Quantity
from automotive_manufacture
group by Supplier_Name;

```

| Result Grid |                      |                | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|----------------------|----------------|--------------|---------|--------------------|
|             | Supplier_Name        | Total_Quantity |              |         |                    |
| ▶           | Prime Components     | 27093          |              |         |                    |
|             | DriveLine Systems    | 13735          |              |         |                    |
|             | Global Motors Parts  | 18757          |              |         |                    |
|             | Metro Auto Supply    | 19811          |              |         |                    |
|             | Elite Automotive     | 18897          |              |         |                    |
|             | SteelTech Components | 20555          |              |         |                    |

#9. Find the total Cost by Region.

```

select Region, sum(`Total_Cost`) AS Total_Region
from automotive_manufacture
group by Region;

```

| Result Grid |               |               | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------------|---------------|--------------|---------|--------------------|
|             | Part_Category | Total_Ordered |              |         |                    |
| ▶           | Braking       | 29030         |              |         |                    |
|             | Transmission  | 10027         |              |         |                    |
|             | Electrical    | 25082         |              |         |                    |
|             | Body          | 36149         |              |         |                    |
|             | Steering      | 15091         |              |         |                    |
|             | Engine        | 39349         |              |         |                    |

#10.Find the number of orders by Warehouse.

```
select `Warehouse`, count(*) AS Number_Orders
from automotive_manufacture
group by `Warehouse`;
```

| Result Grid |        |              | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|--------|--------------|--------------|---------|--------------------|
|             | Region | Total_Region |              |         |                    |
| ▶           | West   | 9541535      |              |         |                    |
|             | South  | 12340895     |              |         |                    |
|             | North  | 9540520      |              |         |                    |
|             | East   | 4582715      |              |         |                    |

#11.Find the average Unit Cost by Material.

```
select Material, avg(Unit_Cost) AS Average_Material
from automotive_manufacture
group by Material;
```

| Result Grid |           |               | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-----------|---------------|--------------|---------|--------------------|
|             | Warehouse | Number_Orders |              |         |                    |
| ▶           | Chennai   | 24            |              |         |                    |
|             | Bengaluru | 37            |              |         |                    |
|             | Gurugram  | 46            |              |         |                    |
|             | Pune      | 43            |              |         |                    |

#12.Find the number of on-time orders.

```
select On_Time, count(*) AS Number_Orders
from automotive_manufacture
group by On_Time;
```

| Result Grid |          |                  | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|----------|------------------|--------------|---------|--------------------|
|             | Material | Average_Material |              |         |                    |
| ▶           | Ceramic  | 239.3750         |              |         |                    |
|             | Steel    | 205.0758         |              |         |                    |
|             | Aluminum | 259.2857         |              |         |                    |
|             | Paper    | 320.7692         |              |         |                    |
|             | Plastic  | 237.2414         |              |         |                    |

#13. Find the number of delayed orders.

```
select count(*) AS Number_Delayed_orders
from automotive_manufacture
where On_Time = 'NO';
```

| Result Grid |         |               | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------|---------------|--------------|---------|--------------------|
|             | On_Time | Number_Orders |              |         |                    |
| ▶           | Yes     | 63            |              |         |                    |
|             | No      | 87            |              |         |                    |

#14. Find the On-Time Delivery %.

```
select avg(case when On_Time = 'Yes'
Then 0.1 else 0 end) * 100 AS On_Time_Delivery_Percentage
from automotive_manufacture;
```

| Result Grid |                       |  | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-----------------------|--|--------------|---------|--------------------|
|             | Number_Delayed_orders |  |              |         |                    |
| ▶           | 87                    |  |              |         |                    |

#15. Find the number of on-time orders by supplier.

```
select Supplier_Name, count(*) AS On_Time_Orders
from automotive_manufacture
where On_Time = 'Yes'
group by Supplier_Name;
```

| Result Grid |                             | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-----------------------------|--------------|---------|--------------------|
|             | On_Time_Delivery_Percentage |              |         |                    |
| ▶           | 4.20000                     |              |         |                    |

#16. Find the number of delayed orders by supplier.

```
select Supplier_Name, count(*) AS Delayed_Orders
from automotive_manufacture
where On_Time = 'No'
group by Supplier_Name;
```

| Result Grid |                      | Filter Rows:   | Export: | Wrap Cell Content: |
|-------------|----------------------|----------------|---------|--------------------|
|             | Supplier_Name        | On_Time_Orders |         |                    |
| ▶           | Prime Components     | 7              |         |                    |
|             | Metro Auto Supply    | 11             |         |                    |
|             | SteelTech Components | 8              |         |                    |
|             | Global Motors Parts  | 9              |         |                    |
|             | Elite Automotive     | 4              |         |                    |

#18. Find suppliers having more than 5 delayed orders.

```
select Supplier_Name, count(*) AS Delayed_Orders
from automotive_manufacture
where On_Time = 'No'
group by Supplier_Name
having count(*) > 5;
```

| Result Grid |                     | Filter Rows:   | Export: | Wrap Cell Content: |
|-------------|---------------------|----------------|---------|--------------------|
|             | Supplier_Name       | Delayed_Orders |         |                    |
| ▶           | DriveLine Systems   | 7              |         |                    |
|             | Global Motors Parts | 12             |         |                    |
|             | Elite Automotive    | 11             |         |                    |
|             | Prime Components    | 15             |         |                    |
|             | Precision Auto      | 15             |         |                    |

#19. Find the supplier with the highest average Lead Time.

```
select Supplier_Name, avg(`Lead_Time_Days`) AS Average_Lead_Time
from automotive_manufacture
```

group by Supplier\_Name

order by Average\_Lead\_Time desc

limit 1;

| Result Grid | Filter Rows:     | Export:           | Wrap Cell Content: |
|-------------|------------------|-------------------|--------------------|
|             | Supplier_Name    | Average_Lead_Time |                    |
| ▶           | Elite Automotive | 16.5333           |                    |

#20.Find orders where Delivered Quantity is less than Ordered Quantity.

select \*

from automotive\_manufacture

where Delivered\_Qty < Ordered\_Qty;

| Order_ID | Supplier_ID | Supplier_Name       | Part_ID | Part_Name    | Part_Category | Material | Order_Date | Required_Date | Delivery_Date | Ordered_Qty | Delivered_Qty | Unit_Cost | Total_Cost | Defect_Qty | Lead_T |
|----------|-------------|---------------------|---------|--------------|---------------|----------|------------|---------------|---------------|-------------|---------------|-----------|------------|------------|--------|
| ▶ O0001  | S002        | Prime Components    | P001    | Brake Pad    | Braking       | Ceramic  | 06-05-2025 | 16-05-2025    | 15-05-2025    | 1608        | 1595          | 275       | 438625     | 22         | 9      |
| O0002    | S007        | DriveLine Systems   | P001    | Brake Pad    | Braking       | Ceramic  | 17-02-2025 | 27-02-2025    | 01-03-2025    | 1134        | 1057          | 45        | 47565      | 71         | 12     |
| O0003    | S004        | Global Motors Parts | P012    | Clutch Plate | Transmission  | Steel    | 23-04-2025 | 07-05-2025    | 16-05-2025    | 669         | 566           | 600       | 339600     | 0          | 23     |
| O0004    | S003        | Metro Auto Supply   | P012    | Clutch Plate | Transmission  | Steel    | 24-06-2025 | 05-07-2025    | 03-07-2025    | 540         | 443           | 110       | 48730      | 6          | 9      |

#21.Find the total shortage quantity.

select sum(Ordered\_Qty - Delivered\_Qty) AS Total\_Shortage\_Quantity

from automotive\_manufacture

where Delivered\_Qty < Ordered\_Qty;

| Order_ID | Supplier_ID | Supplier_Name       | Part_ID | Part_Name    | Part_Category | Material | Order_Date | Required_Date | Delivery_Date | Ordered_Qty | Delivered_Qty | Unit_Cost | Total_Cost | Defect_Qty | Lead_Ti |
|----------|-------------|---------------------|---------|--------------|---------------|----------|------------|---------------|---------------|-------------|---------------|-----------|------------|------------|---------|
| ▶ O0001  | S002        | Prime Components    | P001    | Brake Pad    | Braking       | Ceramic  | 06-05-2025 | 16-05-2025    | 15-05-2025    | 1608        | 1595          | 275       | 438625     | 22         | 9       |
| O0002    | S007        | DriveLine Systems   | P001    | Brake Pad    | Braking       | Ceramic  | 17-02-2025 | 27-02-2025    | 01-03-2025    | 1134        | 1057          | 45        | 47565      | 71         | 12      |
| O0003    | S004        | Global Motors Parts | P012    | Clutch Plate | Transmission  | Steel    | 23-04-2025 | 07-05-2025    | 16-05-2025    | 669         | 566           | 600       | 339600     | 0          | 23      |
| O0004    | S003        | Metro Auto Supply   | P012    | Clutch Plate | Transmission  | Steel    | 24-06-2025 | 05-07-2025    | 03-07-2025    | 540         | 443           | 110       | 48730      | 6          | 9       |

#22.Find the shortage quantity by supplier.

select Supplier\_Name,sum(Ordered\_Qty - Delivered\_Qty) AS Shortage\_quantity

from automotive\_manufacture

where Delivered\_Qty < Ordered\_Qty

group by Supplier\_Name;

| Result Grid |                     |                   | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------------------|-------------------|--------------|---------|--------------------|
|             | Supplier_Name       | Shortage_quantity |              |         |                    |
| ▶           | Prime Components    | 1115              |              |         |                    |
|             | DriveLine Systems   | 846               |              |         |                    |
|             | Global Motors Parts | 1289              |              |         |                    |
|             | Metro Auto Supply   | 1520              |              |         |                    |
|             | Elite Automotive    | 718               |              |         |                    |

#23.What is the On-Time Delivery %?

```
select avg(case
when On_Time = 'Yes' Then 1.0
else 0 end) * 100 AS On_Time_Delivery_Percentage
from automotive_manufacture;
```

| Result Grid |                             |  | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-----------------------------|--|--------------|---------|--------------------|
|             | On_Time_Delivery_Percentage |  |              |         |                    |
| ▶           | 42.00000                    |  |              |         |                    |

#24.What is the Defect Rate %?

```
SELECT
(SUM(Defect_Qty) / SUM(Delivered_Qty)) * 100 AS Defect_Percentage
FROM automotive_manufacture;
```

| Result Grid |                   |  | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-------------------|--|--------------|---------|--------------------|
|             | Defect_Percentage |  |              |         |                    |
| ▶           | 4.0639            |  |              |         |                    |

#25.Find suppliers with both high defects and high lead time.

```
select Supplier_Name, sum(` Defect_Qty` ) AS Total_Defect_Qty,
Avg(` Lead_Time_Days` ) AS Avg_Lead_Time_Days
from automotive_manufacture
group by Supplier_Name
having sum(` Defect_Qty` ) > 100 and
```

avg(`Lead\_Time\_Days`) > 15;

| Result Grid         | Filter Rows:     | Export:            | Wrap Cell Content: |
|---------------------|------------------|--------------------|--------------------|
| Supplier_Name       | Total_Defect_Qty | Avg_Lead_Time_Days |                    |
| Prime Components    | 918              | 15.6818            |                    |
| Global Motors Parts | 617              | 15.6190            |                    |
| Elite Automotive    | 620              | 16.5333            |                    |
| Precision Auto      | 1028             | 15.7391            |                    |
| AutoParts India     | 873              | 15.7778            |                    |

#26.Combine Orders with Supplier Contact Details

```
select a.Order_ID ,  
s.Supplier_Name,  
s.Contact_Person  
from automotive_manufacture a  
inner join supplier_details s  
on a.Supplier_ID = s.Supplier_ID;
```

| Result Grid         | Filter Rows:     | Export:            | Wrap Cell Content: |
|---------------------|------------------|--------------------|--------------------|
| Supplier_Name       | Total_Defect_Qty | Avg_Lead_Time_Days |                    |
| Prime Components    | 918              | 15.6818            |                    |
| Global Motors Parts | 617              | 15.6190            |                    |
| Elite Automotive    | 620              | 16.5333            |                    |
| Precision Auto      | 1028             | 15.7391            |                    |
| AutoParts India     | 873              | 15.7778            |                    |

#27.Find all orders supplied by suppliers located in Bengaluru.

```
select a.Order_ID,  
a.Supplier_ID,  
s.Supplier_Name  
from automotive_manufacture a  
join supplier_details s  
on a.Supplier_ID = s.Supplier_ID  
where s.Supplier_Location = "Bengaluru";
```

| Result Grid |          |              |                    |                                                                                                                                                                                                   |
|-------------|----------|--------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |          | Filter Rows: |                    | Export:  Wrap Cell Content:  |
|             | Order_ID | Supplier_ID  | Supplier_Name      | Supplier_Location                                                                                                                                                                                 |
| ▶           | O0007    | S006         | AutoTech Solutions | Bengaluru                                                                                                                                                                                         |
|             | O0009    | S006         | AutoTech Solutions | Bengaluru                                                                                                                                                                                         |
|             | O0022    | S006         | AutoTech Solutions | Bengaluru                                                                                                                                                                                         |
|             | O0029    | S001         | ABC Auto Parts     | Bengaluru                                                                                                                                                                                         |
|             | O0030    | S001         | ABC Auto Parts     | Bengaluru                                                                                                                                                                                         |

#28. Find suppliers who have On\_Time = 'No'.

```
select distinct a.Supplier_Name
from automotive_manufacture a
join supplier_details s
on a.Supplier_ID = s.Supplier_ID
where On_Time = 'No';
```

Result Grid




Filter Rows:

Export:



Wrap Cell Content:



|   | Supplier_Name       |
|---|---------------------|
| ▶ | DriveLine Systems   |
|   | Global Motors Parts |
|   | Elite Automotive    |
|   | Prime Components    |
|   | Precision Auto      |

#29. Display all suppliers and their corresponding manufacturing records, including suppliers who have no manufacturing records.

```
select s.Supplier_Name,
s.Supplier_ID,
a.Order_ID
from automotive_manufacture a
left join supplier_details s
on a.Supplier_ID = s.Supplier_ID ;
```

| Result Grid |                  |             |          | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|------------------|-------------|----------|--------------|---------|--------------------|
|             | Supplier_Name    | Supplier_ID | Order_ID |              |         |                    |
| ▶           | Prime Components | S002        | O0001    |              |         |                    |
|             | Precision Parts  | S007        | O0002    |              |         |                    |
|             | SteelWorks Ltd   | S004        | O0003    |              |         |                    |
|             | Global Motors    | S003        | O0004    |              |         |                    |
|             | Prime Components | S002        | O0005    |              |         |                    |

#30. Identify Suppliers Without Any Orders

```
select s.Supplier_ID,
s.Supplier_Name,
s.Supplier_Location,
s.Contact_Person
from automotive_manufacture a
right join supplier_details s
on a.Supplier_ID = s.Supplier_ID;
```

| Result Grid |             |                |                   |                | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-------------|----------------|-------------------|----------------|--------------|---------|--------------------|
|             | Supplier_ID | Supplier_Name  | Supplier_Location | Contact_Person |              |         |                    |
| ▶           | S001        | ABC Auto Parts | Bengaluru         | Ravi           |              |         |                    |
|             | S001        | ABC Auto Parts | Bengaluru         | Ravi           |              |         |                    |
|             | S001        | ABC Auto Parts | Bengaluru         | Ravi           |              |         |                    |
|             | S001        | ABC Auto Parts | Bengaluru         | Ravi           |              |         |                    |
|             | S001        | ABC Auto Parts | Bengaluru         | Ravi           |              |         |                    |

#31. Combine two result sets containing Supplier\_ID and Order\_ID using UNION.

```
select Supplier_ID ,Order_ID
from automotive_manufacture
union
select Supplier_ID , Order_ID
from automotive_manufacture ;
```

|             |                                                                                   |                                                                                                                     |                                                                                            |                                                                                                        |
|-------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Result Grid |  |  Filter Rows: <input type="text"/> | Export:  | Wrap Cell Content:  |
|             | Supplier_ID                                                                       |                                                                                                                     |                                                                                            |                                                                                                        |
| ▶           | S002                                                                              |                                                                                                                     |                                                                                            |                                                                                                        |
|             | S007                                                                              |                                                                                                                     |                                                                                            |                                                                                                        |
|             | S004                                                                              |                                                                                                                     |                                                                                            |                                                                                                        |
|             | S003                                                                              |                                                                                                                     |                                                                                            |                                                                                                        |
|             | S008                                                                              |                                                                                                                     |                                                                                            |                                                                                                        |